

The Adolescent Scar of Early-Life Disaster

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Tables and instrument appendix — draft for discussion

OVERVIEW

The 27 February 2010 earthquake, of magnitude 8.8, struck a large part of Chile. Fourteen years later the ELPI, Chile's early-childhood panel, returned to children who were between six months and four years old that day and, for the first time, asked the adolescents themselves about their mental health. We find that the earthquake left a scar whose depth depends on the age at which it struck. Adolescents who lived through it at age two report fewer anxiety and depression symptoms at ages 15–18 than adolescents struck as infants, when attachment and stress regulation are forming, or at ages three and four, when children form their first lasting memories, a U-shaped profile whose trough is consistent with the window of infantile amnesia (Terr 1988; Bauer 2007; Alberini and Travaglia 2017). Girls explain between 81 and 95 percent of this profile, and their caregivers do not perceive these problems, reporting no gradient at any age.

Beyond mental health, the 2024 wave lets us follow the developmental damage that Gillmore (2026) documents after the same earthquake. His cognitive result survives the passage to adolescence, since the vocabulary loss he measures at ages 7–11, of 0.17 standard deviations, is still there at ages 15–18, at 0.24. His non-cognitive result does not survive it, since the caregiver-reported behavior problems he finds at ages 7–11 are gone at ages 15–18 (Table 2). By contrast, the mental-health effects documented in this paper appear in the symptoms that adolescents report about themselves, while for the same adolescents the caregiver reports used by Gillmore (2026) show no effect.

The magnitude is moderate. Measured with the PHQ-4, a four-question screener of low mood, loss of interest, nervousness and worry that the adolescents answer in private, exposure at age two lowers the score by 0.39 points on its 0-to-12 scale relative to exposure before age one, or 0.147 standard deviations, statistically significant at the 5 percent level (column 3 of Panel A in Table 3). The index averages 4.31 points in the sample, so the gap corresponds to 9 percent of the mean and to about a third of the 1.11-point distance between girls and boys, and the distance by mother's education is essentially zero. The anxiety half of the index moves the most, since the probability of a positive GAD-2 screen falls by 7.3 percentage points relative to a base of 27.3 percent.

Among girls the PHQ-4 gap doubles to 0.300 standard deviations and appears in both halves of the index, with a fall of 13.7 percentage points in the anxiety screen and of 11.7 in the depression screen, while among boys every estimate is close to zero and statistically insignificant (Table 5). Among caregivers the same design finds nothing at any age (Table 4), which is consistent with symptoms that adolescents feel and adults do not observe. Figure 1 draws the full age profile.

EMPIRICAL STRATEGY

Every adolescent of the 2024 wave lived through the earthquake as a child, and identification rests on the two things that vary across them, how hard their municipality was shaken and how old they were when it struck, with age grouped as 0–11 months (the omitted group), 12–23, 24–35 and 36–59. We compare the outcome gap between two exposure ages within the municipalities the earthquake hit against the same gap within the municipalities it did not hit, so municipality fixed effects keep every comparison local and birth-cohort fixed effects absorb everything a cohort shares, including its age at the survey. All tables estimate versions of

$$Y_{imc} = \alpha + \sum_a \beta_a (\text{Bin}_{ai} \times D_m) + \sum_a \gamma_a \text{Bin}_{ai} + \psi_m + \theta_c + X_i' \delta + \varepsilon_{imc}, \quad (1)$$

where Y_{imc} is the outcome of adolescent i of birth cohort c in municipality m ; Bin_{ai} are the exposure-age groups; D_m equals one in the six regions where the official intensity study records destructive shaking (V to IX and the Metropolitan Region; Astroza et al. 2010); ψ_m and θ_c are municipality-of-selection and birth-cohort fixed effects; X_i are household controls; and ε_{imc} is the error term. The coefficients of interest are the β_a , the effect of having lived the earthquake in an affected municipality at age a relative to the children exposed before age one. The main effects γ_a absorb the outcome differences across exposure-age groups that are common to affected and unaffected municipalities, differences that exist mechanically because a child who was older at the earthquake is also older at the 2024 interview, and β_a is therefore identified only from how that age profile differs between the two zones. Standard errors are clustered by municipality of selection, and estimates use the 2024 sampling weights.

Table 1
Descriptive statistics.

	(1) Non-Earthquake	(2) Earthquake	(3) Difference
Girl	0.489 (0.500)	0.489 (0.500)	−0.000 (0.011)
Age at the earthquake (months)	26.930 (12.710)	27.098 (12.614)	0.169 (0.254)
Mother's age (2010)	28.980 (7.066)	29.676 (7.116)	0.695*** (0.196)
Mother's schooling (2010)	10.312 (3.182)	10.311 (3.180)	−0.002 (0.211)
HH size (2010)	4.865 (1.641)	4.948 (1.686)	0.083 (0.074)
Rural household (2010)	0.080 (0.271)	0.094 (0.292)	0.014 (0.035)
Children of the mother (2012)	2.165 (1.072)	2.155 (1.075)	−0.010 (0.029)
Prior mental health, mother (2012)	0.091 (0.288)	0.125 (0.331)	0.034*** (0.012)
Prior mental health, father (2012)	0.023 (0.150)	0.027 (0.161)	0.004 (0.004)
Prior mental health, relative (2012)	0.123 (0.328)	0.152 (0.359)	0.029* (0.017)
Observations	2,349	7,654	10,003

Notes: ELPI uses sampling weight at national level. The sample are the 10,003 adolescents of the 2024 wave, born January 2006–August 2009, aged 6 to 49 months on the day of the earthquake and 15 to 18 at the 2024 interview. Sex and ages are fixed at birth and the 2010 rows come from the pre-earthquake baseline; the 2012 rows are the family size and the lifetime conditions reported in the 2012 wave, the controls of the main specification, and are missing for adolescents without that interview. Mental health rows mark a reported diagnosis of depression, anxiety, bipolar disorder, schizophrenia, a personality disorder or an autism spectrum disorder. Earthquake municipalities are those of the six regions where the official intensity study records destructive shaking (V, VI, VII, VIII, IX and Metropolitan; Astroza et al. 2010). Column 3 is the difference from a regression on the earthquake indicator; standard errors clustered at the municipality of selection appear in parentheses—number of clusters: 116. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 2

The impact of the earthquake on cognitive and non-cognitive development in the medium- and long-term.

	Medium-term effect (Gillmore) 2017 wave		Long-term effect 2024 wave	
	(1)	(2)	(3)	(4)
<i>Panel A: Peabody</i>				
Affected×Earthquake	−0.174** (0.0686)	−0.280*** (0.0866)	−0.240** (0.104)	−0.412** (0.180)
Observations	14,069	14,069	13,779	13,779
<i>Panel B: CBCL2</i>				
Affected×Earthquake	−0.129* (0.0763)	−0.220** (0.105)	0.084 (0.0884)	−0.066 (0.125)
Observations	11,568	11,568	11,038	11,038
Municipality FE	✓	✓	✓	✓
Birth Cohort FE	✓	✓	✓	✓
Child-Mother-HH control	✓	✓	✓	✓
Municipality linear trends	×	✓	×	✓

Notes: Each column represents a separate regression. The dependent variable is the test score measured in standard deviations. Peabody is the Peabody Test Score and CBCL is the Child Behavior Check List measure, oriented so that higher means fewer problems. Columns 1 and 2 reproduce Columns 3 and 4 of Table 3 in Gillmore (2026): children exposed between in utero and age four, with scores measured seven years later, at ages 7–11; our replication of his Column 3 on the public ELPI files gives −0.160 for Peabody and −0.103 for CBCL2. Columns 3 and 4 estimate the same two specifications on the 2024 wave: the same exposed children measured fourteen years later, at ages 15–18, against the comparison group conceived after the earthquake and measured in 2017; pooling the two waves adds survey-wave fixed effects, as in Gillmore’s short-term table. Standard errors clustered at the municipal level appear in parentheses—number of clusters: 145 in Columns 1 and 2 and 116 in Columns 3 and 4. The estimates use sampling weights. Child/Mother/HH characteristics include gender, age at the moment of the test (in months), birth order, mothers’ age and schooling, father present, number of household members, prior mental health condition for mother, father and relative. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

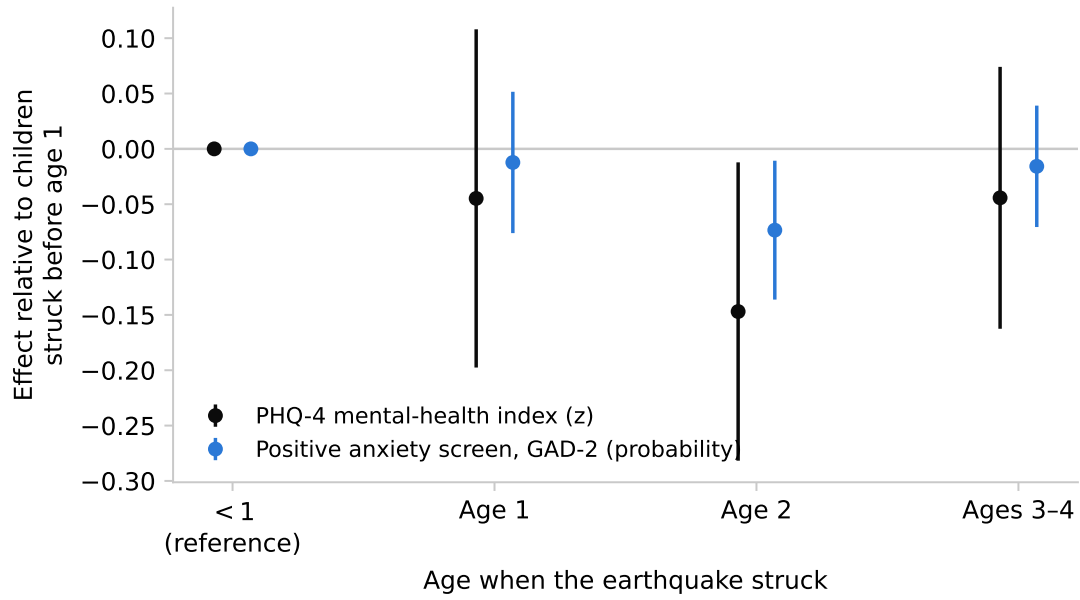


Figure 1. Estimates by Age at Exposure

Notes: Each marker is a β_a coefficient of equation (1) with the controls of column 3 of Table 3, with its 95 percent confidence interval. The reference group, drawn at zero by construction, are the children struck at 0–11 months — already born; children in utero during the earthquake are not in the 2024 wave, which follows only the cohort born 2006–2009. The PHQ-4 is in standard deviations and the GAD-2 screen in probability units. Standard errors clustered by municipality of selection in parentheses.

Table 3

Age at Exposure to the Earthquake and Self-Reported Mental Health at Ages 15–18

	(1)	(2)	(3)	(4)
<i>Panel A. Mental health: PHQ-4 index, four questions on low mood, loss of interest, nervousness and worry (0–12, z)</i>				
Exposed at age 1	–0.051 (0.080)	–0.044 (0.078)	–0.045 (0.078)	–0.030 (0.079)
Exposed at age 2	–0.144** (0.072)	–0.153** (0.069)	–0.147** (0.069)	–0.120 (0.076)
Exposed at ages 3–4	–0.038 (0.064)	–0.048 (0.060)	–0.044 (0.060)	–0.011 (0.063)
<i>Panel B. The anxiety half of the index: positive GAD-2 screen (its two anxiety questions, subscore of 3 or more)</i>				
Exposed at age 1	–0.014 (0.032)	–0.012 (0.032)	–0.012 (0.033)	–0.018 (0.035)
Exposed at age 2	–0.073** (0.032)	–0.076** (0.032)	–0.073** (0.032)	–0.064* (0.035)
Exposed at ages 3–4	–0.015 (0.027)	–0.018 (0.028)	–0.016 (0.028)	–0.015 (0.030)
<i>Panel C. The depression half of the index: positive PHQ-2 screen (its two depression questions, subscore of 3 or more)</i>				
Exposed at age 1	–0.017 (0.039)	–0.014 (0.038)	–0.014 (0.038)	–0.020 (0.041)
Exposed at age 2	–0.046 (0.042)	–0.050 (0.041)	–0.048 (0.041)	–0.073 (0.045)
Exposed at ages 3–4	–0.008 (0.032)	–0.011 (0.031)	–0.010 (0.031)	–0.017 (0.035)
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	No	Yes	Yes	Yes
2012 controls	No	No	Yes	Yes
Excludes the Metropolitan Region	No	No	No	Yes
Observations	9,996	9,996	9,996	6,430
Selection municipalities (clusters)	116	116	116	67

Notes: Ordinary least squares estimates of equation (1) on the adolescents of the 2024 ELPI wave; seven of the 10,003 adolescents of Table 1 drop for incomplete PHQ-4 answers. Panels B and C disaggregate the Panel A index into its two pairs of questions, expressed as clinical screens; Appendix Table A1 lists every question verbatim. Exposure-age groups give the child's age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. The affected zone comprises the six regions where the official intensity study records destructive shaking (V, VI, VII, VIII, IX and Metropolitan; Astroza et al. 2010). Column 1 includes exposure-age, municipality-of-selection and birth-cohort fixed effects; column 2 adds pre-earthquake controls (sex, age in months, and the 2010 household baseline: mother's education and age, household size, rurality); column 3 adds 2012 controls (number of children of the mother, family history of mental illness), with missing-value indicators; column 4 estimates column 3 without the Metropolitan Region. Estimates use the 2024 sampling weights. The sample is the same in every panel; column 4 has 6,430 adolescents in 67 municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 4
Caregiver-Reported Outcomes

	(1)	(2)	(3)	(4)
<i>Panel A. CBCL internalizing score in 2024: caregiver checklist (T, z)</i>				
Exposed at age 1	−0.057 (0.099)	−0.047 (0.096)	−0.055 (0.095)	−0.024 (0.100)
Exposed at age 2	−0.121 (0.074)	−0.121 (0.075)	−0.109 (0.075)	−0.131 (0.084)
Exposed at ages 3–4	−0.086 (0.069)	−0.084 (0.068)	−0.077 (0.069)	−0.069 (0.074)
Observations	9,970	9,970	9,970	6,412
<i>Panel B. Within-child change in the same CBCL score, 2017 to 2024 (z by wave)</i>				
Exposed at age 1	−0.030 (0.131)	−0.011 (0.125)	−0.012 (0.124)	−0.020 (0.128)
Exposed at age 2	−0.009 (0.104)	−0.004 (0.102)	−0.002 (0.102)	−0.045 (0.110)
Exposed at ages 3–4	−0.040 (0.109)	−0.049 (0.106)	−0.050 (0.106)	−0.093 (0.112)
Observations	7,169	7,169	7,169	4,901
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	No	Yes	Yes	Yes
2012 controls	No	No	Yes	Yes
Excludes the Metropolitan Region	No	No	No	Yes
Selection municipalities (clusters)	116	116	116	67

Notes: Same design and specifications as Table 3, with the caregiver’s report of the child as the outcome. The CBCL internalizing score is reported by the main caregiver, in T units, standardized within age. Panel B uses the change between the 2017 measurement (ages 8–12) and the 2024 measurement (ages 15–18) of the same child, each standardized within its wave and age; with two periods this is equivalent to a child fixed-effects estimate. Neither panel shows an exposure-age gradient: the scar of Table 3 appears only in what adolescents report in private. Exposure-age groups give the child’s age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Column 1 includes exposure-age, municipality-of-selection and birth-cohort fixed effects; column 2 adds pre-earthquake controls (sex, age in months, and the 2010 household baseline: mother’s education and age, household size, rurality); column 3 adds 2012 controls (number of children of the mother, family history of mental illness), with missing-value indicators; column 4 estimates column 3 without the Metropolitan Region. Estimates use the 2024 sampling weights. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

Table 5

The effect of exposure at age two on adolescent mental health, by sex, mother's education and area of residence.

	Sex		Mother's education		Household in 2010	
	Girls (1)	Boys (2)	Secondary (3)	Tertiary (4)	Urban (5)	Rural (6)
Panel A: PHQ-4 index						
Exposed at age 2	−0.300*** (0.100)	−0.016 (0.098)	−0.107 (0.074)	−0.201 (0.188)	−0.139* (0.076)	−0.238 (0.172)
Observations	4,961	5,035	7,755	2,053	9,005	991
Panel B: Anxiety, positive GAD-2 screen						
Exposed at age 2	−0.137*** (0.047)	−0.026 (0.048)	−0.082** (0.034)	−0.021 (0.098)	−0.068** (0.034)	−0.113 (0.092)
Observations	4,961	5,035	7,755	2,053	9,005	991
Panel C: Depression, positive PHQ-2 screen						
Exposed at age 2	−0.117** (0.057)	0.016 (0.045)	−0.030 (0.043)	−0.082 (0.089)	−0.046 (0.045)	−0.073 (0.084)
Observations	4,961	5,035	7,755	2,053	9,005	991
Municipality FE	✓	✓	✓	✓	✓	✓
Birth Cohort FE	✓	✓	✓	✓	✓	✓
Pre-earthquake controls	✓	✓	✓	✓	✓	✓
2012 controls	✓	✓	✓	✓	✓	✓

Notes: Each column represents a separate regression of equation (1) with the controls of column 3 of Table 3, estimated on the subsample named in the column heading. The coefficient reported is the effect of exposure at age two in an affected municipality, relative to exposure at 0–11 months. The PHQ-4 index is measured in standard deviations; the GAD-2 and PHQ-2 screens equal one for a subscore of three or more on the corresponding pair of questions. Mother's education is measured in 2010, and secondary means twelve years of schooling or less. Weighting girls and boys by their shares of the sample, girls account for 95 percent of the age-two coefficient and for 81 percent of the gap between exposure at ages 3–4 and at age two. The PHQ-4 index averages 4.88 points for girls and 3.76 for boys. Standard errors clustered by municipality of selection appear in parentheses. The estimates use sampling weights. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

APPENDIX. INSTRUMENTS: EVERY ITEM AND ITS SCALE

Table A1
Mental-Health Instruments: Items and Scales

	Question, verbatim (1)	Response scale (2)
<i>Panel A. PHQ-4, answered by the adolescent in private (2024 wave)</i>		
d4_1	¿Te has sentido bajoneado(a), deprimido(a), irritable o desesperanzado(a)?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
d4_2	¿Has sentido poco interés o placer al hacer las cosas?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
d4_3	¿Te has sentido muy nervioso(a), angustiado(a) o con los nervios de punta?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
d4_4	¿No has podido dejar de preocuparte o controlar la preocupación?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
<i>Panel B. Caregiver-reported behavior</i>		
cbc12	Child Behavior Checklist, internalizing scale, answered by the main caregiver about the adolescent; the survey releases the computed T score, not the copyrighted items.	T score, standardized in the analysis

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. The PHQ-4 score used in the paper is the sum of the four items rescaled to 0–3 each (0–12 in total) and standardized within age in years. The positive PHQ-2 (depression) and GAD-2 (anxiety) screens are the survey-provided indicators for a subscore of three or more on the corresponding pair of items.